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Energy Conversion

Assignment 1

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1 Problem 1

1.1 a)

The equivalent circuit has four reluctance (one for each leg) and a MMF. Reluctance is given by following equation.

$$R = \frac{l_{mean}}{\mu A} \quad (1)$$

Values are given below.

$$\begin{aligned} R_L &= \frac{22}{\mu 120 \text{ cm}} & R_T &= \frac{24.5}{\mu 100 \text{ cm}} \\ R_R &= \frac{22}{\mu 70 \text{ cm}} & R_B &= \frac{24.5}{\mu 100 \text{ cm}} \end{aligned}$$

MMF is like a voltage source in the electrical circuits and reluctance acts like a resistor. Hence,

$$\varphi = \frac{Ni}{R_{eq}} \rightarrow i = \varphi \frac{R_{eq}}{N} = 629 \text{ mA} \quad (2)$$

1.2 b)

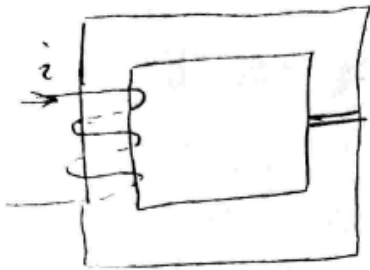
Magnetic flux is constant in the core (assuming that there is no flux leakage). So, the flux density is given by

$$\rho = \frac{\varphi}{A}. \quad (3)$$

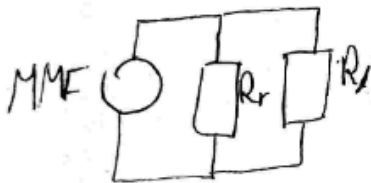
Hence,

$$\rho_T = 0.8 \text{ Wb/m}^2 \quad , \quad \rho_R = 1.14 \text{ Wb/m}^2. \quad (4)$$

2 Problem 2



$$R = \frac{l_{\text{mean}}}{\mu A_c} + \frac{l_{\text{gap}}}{\mu_0 A_{\text{gap}}}$$



$$R_r = 174.85 \times 10^3$$

$$R_l = 198.53 \times 10^3$$

$$\mathcal{G} = F \left(\frac{1}{R_r} + \frac{1}{R_l} \right) = 7.26 \text{ mWb}$$

$$\mathcal{G}_r = 3.86 \text{ mWb}$$

$$\mathcal{G}_l = 3.40 \text{ mWb}$$

b)

$$B_r = \frac{\mathcal{G}_r}{A_{\text{gap}}} = 0.57 \text{ Wb/m}^2$$

$$B_l = \frac{\mathcal{G}_l}{A_{\text{gap}}} = 0.56 \text{ Wb/m}^2$$

3 Problem 3

We model this problem with a voltage source and a resistor. The magnetic flux is trapped in the core. So, it is constant. The reluctance of the core is obtained by the following equation.

$$R = \frac{l_{mean}}{\mu A} = \frac{320}{\mu 400 \text{ cm}} \quad (5)$$

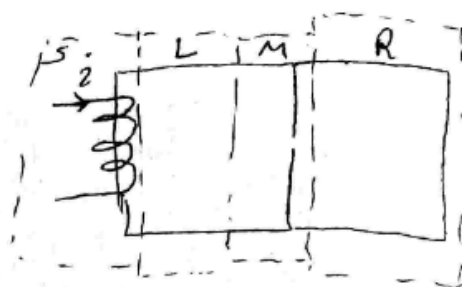
We assume the positive direction of the flux is clockwise.

$$F = N_1 i_1 - N_2 i_2 = 28 \text{ A} \quad (6)$$

$$\varphi = \frac{F}{R} = 8.80 \times 10^{-4} \text{ Wb} \quad (7)$$

4 Problem 4

$$R_{\text{eff}} = (R_L \parallel R_M + R_L) \parallel R_s$$



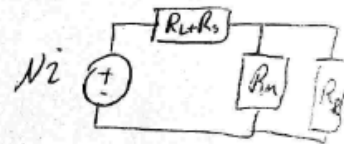
$$R_R = \frac{46}{\mu 120 \text{ cm}}$$

$$R_L = \frac{92}{\mu 120 \text{ cm}}$$

$$R_M = \frac{42}{\mu 200 \text{ cm}} + \frac{0.06}{\mu_0 (1.04) 200 \text{ cm}}$$

$$R_s = \frac{42}{\mu 120 \text{ cm}}$$

$$\mathcal{G}_s = \frac{Ni}{R_{\text{eff}}} = 0.013 \text{ Wb}$$



$$\mathcal{G}_M = \frac{Ni}{R_{\text{eff}}} \frac{R_R}{R_M + R_R} = 4.67 \text{ mWb}$$

$$\mathcal{G}_R = \frac{Ni}{R_{\text{eff}}} \frac{R_M}{R_M + R_R} = 7.83 \text{ mWb}$$

b)

$$B_s = \frac{\mathcal{G}_s}{A_s} = 1.04 \text{ Wb/m}^2$$

$$B_M = \frac{\mathcal{G}_M}{A_M} = 0.2334 \text{ Wb/m}^2$$

$$B_R = \frac{\mathcal{G}_R}{A_R} = 0.653 \text{ Wb/m}^2$$

5 Problem 5

In a core(with no branches), magnetic flux is constant. The flux density is almost constant too(a negligible difference in the air gap). But H is not, it is much smaller in the core than the air gap.

$$\int_s \mathbf{H} \cdot d\mathbf{l} = Ni \quad (8)$$

$$H_c l_c + H_a l_a = Ni \quad (9)$$

In the air gap we have $H_a = \frac{B}{\mu_0}$. Hence,

$$H_c(B) l_c + \frac{B}{\mu_0} l_a = Ni \quad (10)$$

We have to find the intersection point of a line and the curve.

$$B \approx 0.47 T \quad \rightarrow \quad H_a = 374 \times 10^3 \quad (11)$$

6 Problem 6

6.1 a)

As determined before,

$$H = \frac{472 A}{320 cm} = 147.5 \text{ Am}^{-1} \quad (12)$$

$$B = 0.15 \text{ Wb} \quad (13)$$

6.2 b)

$$\mu_r = \frac{H}{B\mu_0} = 809 \quad (14)$$

6.3 c)

It depends on our definition of "good assumption". The assumption has 20 percent error and represent a significant deviation. In general, this is considered as a very large error. So, it is better to simply say that *this is not a good assumption*.

7 Problem 7

Nature opposes changing magnetic flux, according to Faraday's Law. Therefore, the induced voltage always opposes the direction of the flux change.

$$v_{ind} = -N \frac{d\varphi}{dt} \quad (15)$$

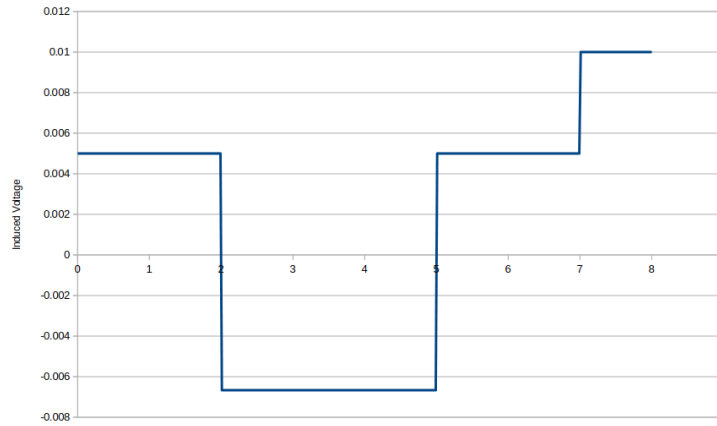


Figure 1: Induced voltage, Problem 7

8 Problem 8

8.1 a)

Knee point is where the B-H graph starts to flatten out. This is where B has the maximum reasonable value. If the core becomes too saturated, increasing the magnetizing force no longer produces much increase in flux density, making the motor inefficient.

$$B \approx 1.3 T \quad \& \quad \mu_r = 3300 \quad (16)$$

8.2 b)

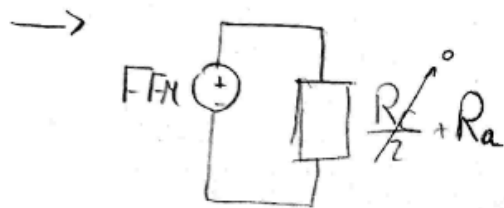
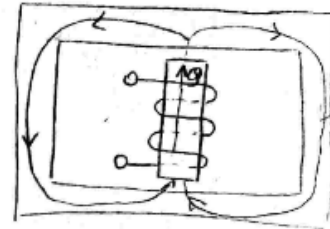
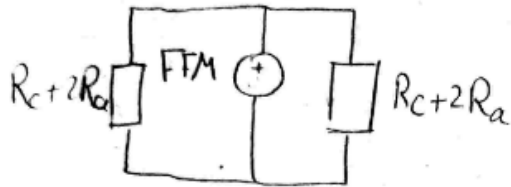
$$\varphi = B_c A_c = 2.08 \times 10^{-3} \text{ Wb} \quad (17)$$

8.3 c)

$$N = \frac{\varphi}{\mu_0 i} \left(\frac{l_c + 2R_{rotor}}{\mu_r A_c} + \frac{l_a}{A_a} \right) \approx 1000 \quad (18)$$

9 Problem 9

(a)



$$R_c \xrightarrow{\mu_r \rightarrow \infty} 0$$

$$R_a = \frac{l_g}{\mu_0 A_g}$$

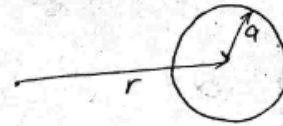
b)

$$B = \frac{\mu_0 N i}{l_g} = 0.075 \text{ Wb/m}^2$$

10 Problem 10

a)

$$B = \mu_0 H = \frac{\mu_0 N i}{2 a r}$$



$$\oint B \cdot da = B(r) \pi a^2 = a^2 \frac{\mu_0 N i}{2 r}$$

$$\rightarrow L = \frac{\mu_0 N^2}{2 r} a^2 = 0.015 \text{ H}$$

b)

$$W = \frac{1}{2\mu_0} \int_{\text{loop}} B^2 d\tau = \frac{1}{2\mu_0} \int_{\text{loop}} B^2 d\tau$$

$$= \frac{\mu_0 N^2 i^2}{2(2\pi r)^2} \int d\tau = \frac{\mu_0 N^2 i^2 a^2}{4r}$$

$\underbrace{\int d\tau}_{= 2\pi r a^2}$

$$W = \frac{1}{2} L i^2 = \frac{\mu_0 N^2 i^2 a^2}{4r}$$

C1

$$i = \alpha t$$

$$\rightarrow \mathcal{E} i = L i \frac{di}{dt} \Rightarrow W = \frac{1}{2} L \alpha^2 T^2$$

$$\rightarrow \alpha = \frac{i}{T} = \frac{\beta(2ar)}{\mu_0 N T}$$

$$\mathcal{E} = \alpha$$